

AIR, WAVES, AND SOUND

- MICRO TO MACRO VIEW OF AIR (THE THING THAT BRINGS US SOUND!)
 - REVIEW BASIC PHYSICS CONCEPTS
 - FUN FACTS

- WAVES
 - TRANSVERSE WAVES (CONCEPTUAL SIMPLICITY)
 - LONGITUDINAL WAVES AND SOUND!

AIR

SOME FAMILIAR PROPERTIES

- TEMPERATURE
- PRESSURE
- SPEED (VELOCITY)
- DENSITY

"THE USUAL" MACRO-SCALE QUANTITIES
FOR DISCUSSING SOUND MECHANICS
(WE WILL IGNORE "ENTROPY", DESPITE HOW DEEPLY
AWESOME IT IS. YOU'RE WELCOME / SORRY / ASK ME)

ALSO

- HUMIDITY
- SMELL
- O_3 CONTENT
- ...

SURVIV



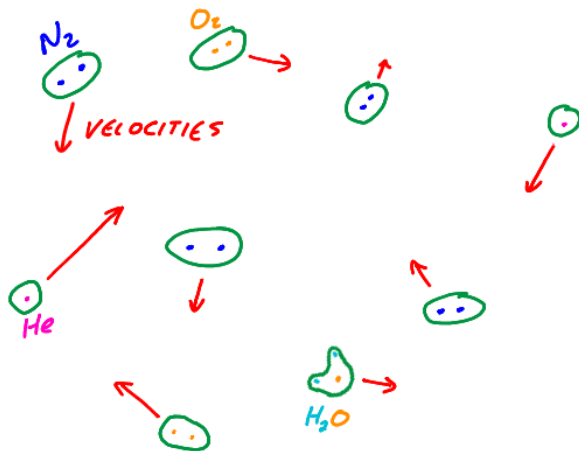
HOW MANY REALLY KNOW THE DIFFERENCE BETWEEN
PRESSURE AND DENSITY?
VELOCITY & SPEED?

10pt



DOES DENSITY AFFECT PRESSURE?

AIR MOLECULES & ATOMS



100pt
AVERAGE / TYPICAL SEPARATION?



10pt
SIZE OF MOLECULES?



20pt
TYPICAL SPEED OF MOLECULE? (IN THIS ROOM)

WALKING	RUNNING	DRIVING	FLYING	$\sim 300 \text{ m/s}$	$\sim 3000 \text{ m/s}$	300000000 m/s
				MACH 1	MACH 10	SPEED OF LIGHT

⇒ AIR MOLECULES ARE TINY, FAST, LOOSELY PACKED, HUGE IN NUMBER!

LEAVE UP

- MONEY QUESTION:

100pt

Q

YOU HAVE WHAT YOU NEED TO ESTIMATE THIS, BUT LET'S GUESS:
HOW FAR DOES A TYPICAL PARTICLE FLY BEFORE IT COLLIDES
WITH ANOTHER?

BONUS

IN A PERFECTLY STILL ROOM, TYPICALLY HOW
LONG DOES IT TAKE FOR A MOLECULE
TO TRAVEL A METER? THE TRUTH MAY
SHOCK YOU. (HINT: "RANDOM WALK")

TEMPERATURE

- CONVERSATIONS THAT DON'T HAPPEN (TO YOU)

"HEY WHAT'S THE AVERAGE MOLECULE SPEED OUTSIDE?"

"WAIT, WHICH MOLECULE? N_2 ?"

"YEAH."

"ABOUT 490 m/s, BUT IT FEELS LIKE 485 WITH THE WIND."

"AH NUTS, WELL, AT LEAST IT MIGHT - WAIT, WHAT
IS 'WIND'? WE HAVEN'T LEARNED ABOUT THAT YET IN
PHYS-224 THE PHYSICS OF MUSIC."

(SHRUGS SHOULDERS) "I'M NOT A PHYSICIST. YOU CAN'T EXPECT
ME TO KNOW WHAT 'WIND' IS."

- INSTEAD WE DISCUSS TEMPERATURE " T ", WHICH CAN HERE BE THOUGHT OF AS "A THING PROPORTIONAL TO THE AVERAGE KINETIC ENERGY OF THE MOLECULES". SPECIFICALLY, IT TURNS OUT

$$\bar{E} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

AVERAGE KINETIC ENERGY → $\bar{E}$
 MASS OF A MOLECULE → m
 AVERAGE OF SQUARED SPEED → $\overline{v^2}$
 SOME CONVERSION CONSTANT ("BOLTZMANN") → k_B

REARRANGING WITH "ALGEBRA"

$$v_{\text{RMS}} = \sqrt{\frac{3k_B T}{m}} \quad \text{OR} \quad T = \frac{m v_{\text{RMS}}^2}{3k_B}$$

TEMPERATURE AND MOLECULE SPEED

BOLTZMANN CONSTANT = $1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}$

NOTE T HERE IS IN KELVINS. SAME SCALE AS CELSIUS BUT SHIFTED SO $T=0$ MEANS "ABSOLUTE ZERO" NOT "WATER FREEZES".

10pt

Q

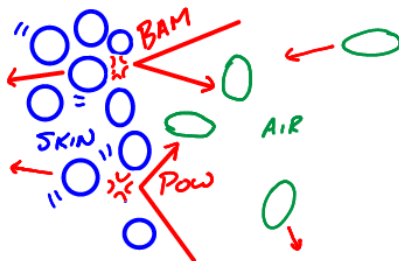
IF THE TEMPERATURE IN THIS ROOM DOUBLES,

(SOPHISTICATED HEAT BEAM)

FUN FACT: IN MY LAB, WE CAN TURN ON A "LASER" THAT COOLS THE MOTION OF A LITTLE PARTICLE TO $\sim 0.001 \text{ K}$

WE ALSO HAVE A FRIDGE THAT CAN COOL A LARGE COPPER PLATE TO 0.01 K .

- SO WHY DOES HOT AIR FEEL HOT ON OUR SKIN?



- MANY TINY BULLETS CAUSING YOUR SKIN ATOMS TO JIGGLE (YOUR SKIN ATOMS ALSO CAN KICK THE AIR, HEATING IT UP!)

WIND VELOCITY

10pt



TYPICAL AVERAGE VELOCITY OF AIR?

"WIND"

← A CHUNK OF AIR,
MADE OF MANY
MOLECULES!

DENSITY

- AIR IS MATTER AND SO IT WEIGHS SOMETHING
 - WHY WE STILL HAVE AN ATMOSPHERE ON EARTH!
 - LOOSELY PACKED, SO IT'S LIGHTER THAN SOLIDS + LIQUIDS
- DENSITY IS SPECIFIED AS MASS PER VOLUME, e.g. kg/m^3

EXAMPLES :

MATERIAL	DENSITY
AIR	1.2 kg/m^3
WOOD	~600 kg/m^3
WATER	1000 kg/m^3
ROCK	~3000 kg/m^3
IRON	7800 kg/m^3

← 1000X DIFFERENCE
CONSISTENT WITH

SEPARATION BEING
ABOUT 10X THE PARTICLE
SIZE. ($10^3 = 1000$)

LOOK! HEY! HERE! → PRESSURE ← MOST IMPORTANT CONCEPT OF THIS LECTURE!!

- PRESSURE IS THE FORCE THAT IS APPLIED PER UNIT AREA: $F = PA$
(MORE AREA ⇒ MORE FORCE.)

FORCE
PRESSURE
AREA

- FAMILIAR PRESSURE IN EARS:

- EARS POPPING ON A PLANE AS THE PRESSURE CHANGES
- HEARING SOUND JUST A FASTER-CHANGING VERSION OF THIS ... WE SHALL SEE!

10pt

Q

A JAR HOLDING VACUUM IS HARDER TO OPEN IF IT IS _____.

- BUT WHAT IS A FORCE?

"FORCE" MEASURED IN "NEWTONS" = kg m/s^2

NEWTON'S LAW:

$$a = \frac{F}{m}$$

NEWTON'S 2nd LAW: PUSHING STUFF ACCELERATES IT

MASS TO ACCELERATE

ACCELERATION:

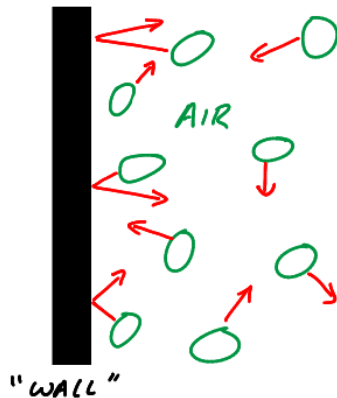
CHANGE IN SPEED PER SECOND ⇒ $\left(\frac{\text{m}}{\text{s}}\right)/\text{s} = \text{m/s}^2$

ALIVE?

Q

HEAVIER THINGS ACCELERATE _____ THAN LIGHTER THINGS UNDER THE SAME FORCE.

- So HOW DOES AIR APPLY FORCE?



- AIR MOLECULES "KICK" SURFACES

- RANDOM TIMES, RANDOM VELOCITIES

- SUMMED TOGETHER, PRODUCES AN AVERAGE FORCE

10pt

Q

AS DRAWN, WHAT WILL HAPPEN TO THE WALL?
(IMAGINE IT IN SPACE)

10pt

Q

WHAT CHANGES IF WE DOUBLE THE PARTICLE DENSITY " N/V "? (N = PARTICLE #, V = GAS VOLUME)

10pt

Q WHAT IF WE SHRINK THE WALL'S AREA "A" BY A FACTOR OF 2?

100pt

Q WHAT IF WE DOUBLE THE TEMPERATURE?

10pt

Q GUESS THE PROPORTIONALITY CONSTANT?

PRESSURE
UNITS: $\frac{N}{m^2}$
= $\frac{kg}{ms^2}$
≡ "PASCAL" (Pa)

1 PASCAL ~ PRESSURE
FROM WEIGHT OF
THICK PAPER

OR, AS MOST WRITE IT,

$$\frac{F}{A} \equiv \boxed{P = \frac{N}{V} k_B T} \quad \text{IDEAL GAS LAW}$$

PRESSURE SCALES WITH DENSITY & TEMPERATURE

- NOTE CHEMISTS DO NOT COUNT ATOMS, RATHER
"MOLES" OF 6.02×10^{23} ATOMS, AND WRITE

$$PV = nRT$$

OF MOLES

SEPARATED VOLUME FROM DENSITY

WHATEVER, "CHEMISTS".

CHEMIST: "LET'S USE QUANTITIES WE CAN HOLD
IN A BEAKER"

PHYSICIST: "SHUT UP, AVOCADO. I'M LIKE LEVITATING
STUFF WITH LASERS RIGHT NOW."

CHEMIST: "YEAH? WELL I CAN MIX TWO CLEAR LIQUIDS
AND GET A RED ONE. OOPS. CANCER."

10pt

Q CAN PRESSURE BE NEGATIVE?

10pt

Q PRESSURE IN THE ROOM RIGHT NOW?

10pt

Q SO I HAVE 100 kN PUSHING ON MY BODY... WHY DON'T I ACCELERATE TO $\sim 1000 \text{ m/s}$ IN 1 SECOND?

10pt

Q WHY DOESN'T MY CHEST COLLAPSE?

- AIR MOLECULES ARE SMALL, BUT THEY ARE FAST AND THERE ARE A LOT OF THEM.

- THUS FAR WE HAVE DISCUSSED STATIC PRESSURE, BUT AIR CAN ALSO CARRY PRESSURE WAVES!

↑ "SOUND"

WAVES

"IF YOU'VE SEEN ONE WAVE YOU'VE SEEN THEM ALL." -RICHARD FEYNMANN

- SOUND IS A WAVE TRAVELING THROUGH AIR

SO WE NEED TO UNDERSTAND WAVES!

↓
SIMPLER TO DRAW &
IMAGINE.

- 2 TYPES OF WAVES WE NEED FOR THIS COURSE: TRANSVERSE & LONGITUDINAL

TRANSVERSE WAVES: A HELPFUL DETOUR ON THE WAY TO SOUND (THAT WE WILL REVISIT)



- MAY OCCUR ON

- A STRING UNDER TENSION (PIANO, GUITAR, VIOLIN, ...)
- THE SURFACE OF WATER
- THROUGHOUT A SOLID
- ELECTROMAGNETIC FIELD
- DRUM HEADS
- ACOUSTIC BODY WALLS

- SINCE THIS KIND OF WAVE IS "INTUITIVE",
LET'S SLOW DOWN AND LOOK AT IT.

PHET DEMO: WAVE ON A STRING (UNDER TENSION)

- NO DAMPING
- NO END

5pt



IF I SHAKE THE WRENCH UP & DOWN
IT WILL SEND WAVES OUT THE WINDOW.
CAN THESE WAVES PUSH THE GREEN BALLS
TOWARD THE WINDOW?

10pt



DO ANY PHYSICAL QUANTITIES MOVE OUT
THE WINDOW AT THE WAVE SPEED?

10pt

Q WHAT DETERMINES THE SPEED OF PROPAGATION?

10pt

Q IF THE BALL MASS INCREASES, WAVE SPEED _____.

10pt

Q IF TENSION INCREASES, WAVE SPEED _____.

DEMO

FOCUS ON ACCELERATION OF 1 BALL...

- TRANSVERSE WAVE EQUATION (QUALITATIVE INTUITION ONLY)

DEMO

OSCILLATE DRIVE + PAUSE

- HOW DO WE MODEL & PREDICT THESE DYNAMICS?

- THE ONLY MOVEMENT IS IN THE "y" (VERTICAL) DIRECTION, SO WE NEED EQUATIONS FOR THE y VALUES AT EACH x

10pt

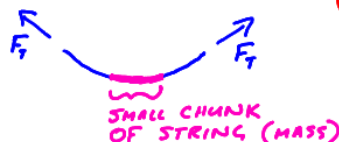
Q WHICH BALLS HAVE THE MOST NET FORCE ON THEM?

10pt

Q WHICH BALLS HAVE THE LEAST NET FORCE?

- FROM THIS, WE MIGHT GUESS

$$(\text{ACCELERATION}) \propto (\text{CURVATURE}) \times (\text{TENSION}) / (\text{MASS})$$



$\propto$ FORCE IN y DIRECTION

- IF YOU'RE CAREFUL (ENTHUSIASTS ONLY, NOT ON TESTS OR HW!)

NEWTON: $a = F / m$

$$\Rightarrow \underbrace{\partial_t^2 y}_{\text{VERTICLE ACCELERATION}} = \underbrace{(\partial_x^2 y)}_{\text{CURVATURE}} \underbrace{F_T / \rho}_{\text{FORCE w/ WHICH WE PULL THE STRING END}} \leftarrow \text{"LINEAR" MASS DENSITY (kg/m)}$$

- PLUGGING IN A TRAVELING WAVE "GUESS"

$$y = y_0 \cos \left[\underbrace{\frac{2\pi f}{c}}_{\text{OSCILLATION FREQUENCY (CYCLES/SECOND) UNITS: "PER SECOND" = HERTZ = Hz}} (x - ct) \right]$$

↑ AMPLITUDE CONSTANT (m) ↑ WAVE SPEED (m/s)

THIS IS "JUST" A COSINE THAT SHIFTS TO THE RIGHT AT SPEED "c"

- DERIVATIVES POP OUT c, COSINES CANCEL, AND WE CAN SOLVE FOR

$$c = \sqrt{\frac{F_T}{\rho}}$$

↑ FORCE STRETCHING THE STRING
↑ MASS PER LENGTH OF THE STRING
↑ TRANSVERSE WAVE SPEED FOR IDEAL STRING
↑ (MENTION MEMBRANE)

10pt
Q

CAN CHANGES IN ρ ALONG THE STRING AFFECT THE FREQUENCY "f" OF THE WAVE?

DEMO

CLAMP THE OTHER END ($\rho \rightarrow \infty$ THERE)
ADD DAMPING, LET IT SETTLE, &
SEE IF THE FREQUENCY IS DIFFERENT ANYWHERE...

- ALSO: FIRST GLIMPSE OF STANDING WAVE RESONANCE!
(HIGH TENSION AROUND 2.07 Hz
NOT LOW ENOUGH DAMPING TO REALLY SEE)
WE WILL COME BACK TO THIS...

MENTION ρ vs. p
↑ GREEK "RHO" FOR DENSITY
↑ USUAL "P" FOR PRESSURE

NOTE THIS "SHAPE" OF EXPRESSION IS COMMON IN WAVES

$$\text{SPEED} \propto \sqrt{\frac{\text{STIFFNESS}}{\text{DENSITY}}}$$

KEEP YOUR EYES PEELED!

(COULD PROBABLY EVEN MODEL "THE WAVE" AT SPORTS EVENTS THIS WAY)

LONGITUDINAL (COMPRESSION) WAVES

- PUSH ONE END OF A STRETCHED SLINKY



- WE SEE OSCILLATIONS IN

- DENSITY
- DISPLACEMENT (BUT PARALLEL TO PROPAGATION)
- TENSION ("PRESSURE"?)

- THESE WAVES OCCUR IN

- AIR ← OOH! OOH! INSTRUMENTS! SOUND!
- LIQUID ← OOH?
- SOLIDS ← OOH!
- PLASMAS
- ELECTRONICS

PHET DEMO WAVES INTRO - SOUND

- PARTICLES, PULSE MODE

- THESE PARTICLES ACT MORE LIKE A SOLID THAN AIR (THEY DON'T COLLIDE/WANDER) BUT IT HELPS VISUALIZE.

10pt

Q IF THE SPEAKER MEMBRANE PULSES, CAN THIS PUSH THE PARTICLES FROM ONE SIDE TO THE OTHER?

DEMO SEND PULSE, SEND ANOTHER & PAUSE

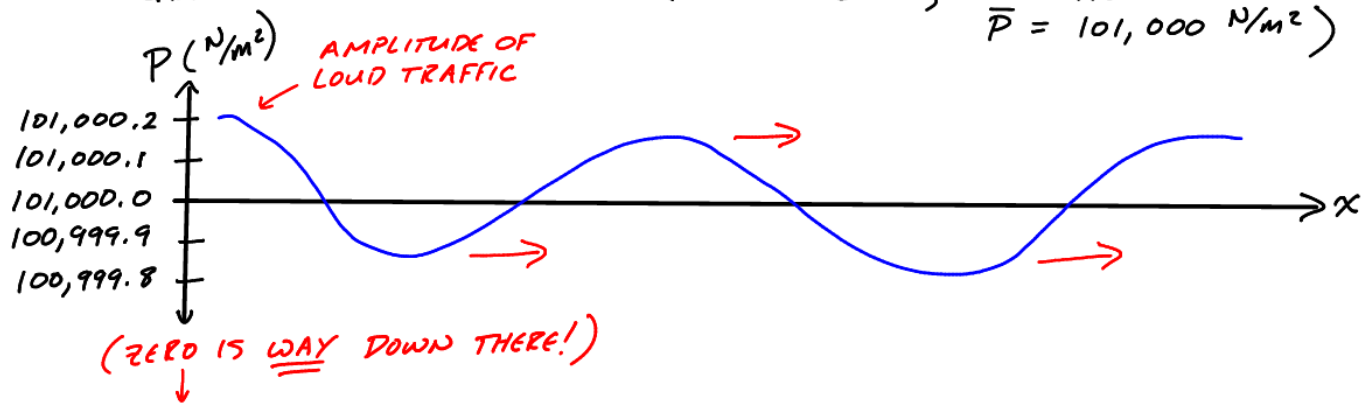
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Q WHERE IS THE PRESSURE (P) GREATEST?
" " " DENSITY (ρ) " ?
" " " VELOCITY (v) " ?
" " " DISPLACEMENT (X) " ?

SHOWS DENSITY
SET "BOTH" AND "SLOW", USE PRESSURE SENSOR AT RED PARTICLE EQUILIBRIUM

DEMO ENABLE GRAPH (NOTE AMPLITUDE FALL-OFF DUE TO ENERGY SPREADING OUT. WE WILL RETURN TO THIS.)

- GRAPHICAL REPRESENTATION (TYPICAL SCALE, WITH MEAN PRESSURE $\bar{P} = 101,000 \text{ N/m}^2$)



BONUS: HOW MUCH DOES AIR DISPLACE DUE TO SOUND?
(SEE ALSO PART 2 BELOW)

- FOLLOWING SIMILAR MATH AS FOR THE STRING (PLUS SOME THERMO!)

1.3 FOR MONATOMIC GAS (He, Ne, Xe, ...)

EQUILIBRIUM (AVERAGE) PRESSURE ($\frac{\text{N}}{\text{m}^2}$)

$$c \approx 1.2 \sqrt{\frac{\bar{P}}{\bar{\rho}}}$$

SPEED OF SOUND IN AIR
VALID FOR DIATOMIC MOLECULES ($\text{N}_2, \text{H}_2, \text{O}_2, \dots$)

EQUILIBRIUM DENSITY IN kg/m^3

ENTHUSIASTS:

$$c = \sqrt{\left. \frac{\partial P}{\partial \rho} \right|_s} = \sqrt{\frac{C_P}{C_V}} \sqrt{\frac{P}{\rho}} \approx 1.2 \sqrt{\frac{P}{\rho}}$$

C_V/P = SPECIFIC HEATS AT
FIXED PRESSURE /
VOLUME

- AT STANDARD TEMPERATURE
AND PRESSURE,

$$c = 344 \text{ m/s} \equiv \text{MACH 1}$$

(~ 1 km IN 3 SEC)



(WHY DO DISTANT BOLTS CRACKLE
& CLOSE ONES "BANG"? :))

20pt

Q WILL THE SPEED OF SOUND DEPEND ON THE GAS (SAME TEMPERATURE)?

- THIS IS A MORE USEFUL FORM IN MY OPINION,
BECAUSE $\bar{P} + \bar{\rho}$ CHANGE TOGETHER $\Rightarrow$ ADDED CONFUSION

10pt

Q DOES c DEPEND ON PRESSURE?

- TO PONDER: - WHY DOES HELIUM MAKE OUR VOICES HIGHER?
- WOULD A HELIUM-FILLED ROOM MAKE A VIOLIN'S PITCH HIGHER? (BEFORE EVERYONE PASSED OUT)
- FLUTE?
- IF A VIOLIN IS PLAYED BY A ROBOT WITH NO EARS IN A ROOM FULL OF PASSED-OUT PEOPLE, DOES IT MAKE A SOUND?
- THAT'S FOR GAS. AGAIN, LOOKS LIKE $\sqrt{\frac{\text{RIGIDITY}}{\text{DENSITY}}}$ $\leftarrow$ HIGHER PRESSURE $\Rightarrow$ HARDER TO COMPRESS.
- FOR LIQUIDS & SOLIDS:

$$c = \sqrt{\frac{1}{\rho \beta_s}}$$

LONGITUDINAL WAVE SPEED
IN LIQUIDS & SOLIDS

"BETA SUB S" = "ADIABATIC COMPRESSIBILITY"

INVERSE RIGIDITY
CHOSEN FOR HISTORICAL
REASONS

GASSES & SLINKIES ARE VERY COMPRESSIBLE COMPARED
TO SOLIDS & LIQUIDS, BUT THEY ALSO HAVE LOWER DENSITY.

GUESS

Q DOES WATER OR AIR HAVE A FASTER WAVE SPEED?

EXAMPLES:

MATERIAL	SOUND SPEED
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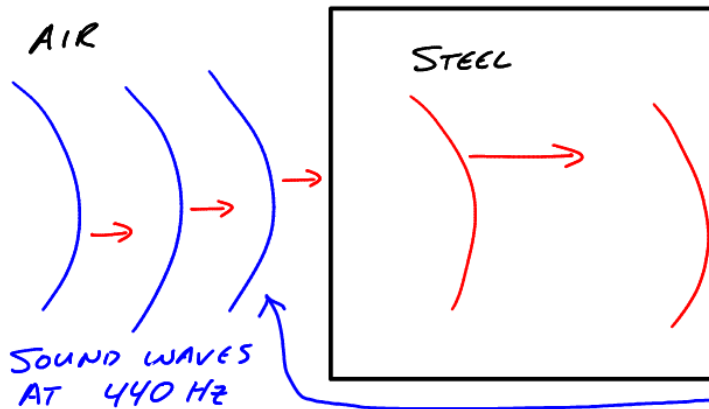
AIR	$\sim 340 \text{ m/s}$
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WATER	1400 m/s
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GLASS	5000 m/s
-------	--------------------

STEEL	5100 m/s
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ALL CONDITION
DEPENDENT TO
SOME DEGREE



10pt
Q

WHAT HAPPENS TO THESE AS THE WAVE ENTERS THE STEEL?

FREQUENCY

SPEED

WAVELENGTH (SPACING OF CRESTS)

- BUT MOST OF THE WAVE WILL REFLECT

- REFLECTIONS GIVE RESONANCES

AND THE ABILITY TO MAKE INSTRUMENTS!!
(MORE LATER)

- FUN FACT: THRESHOLD OF HEARING $\sim 10 \text{ nm/s}$
AIR VELOCITY. TINY!

7 ORDERS OF MAGNITUDE
= HUGE DYNAMIC RANGE!

(14 ORDERS IN POWER!)

MUCH BETTER THAN

ANY MIC + ELECTRONICS

I HAVE FOUND...

SENSITIVE!

THRESHOLD OF PAIN: $\sim 0.1 \text{ m/s}$

BONUS (CONT'D)

THRESHOLDS IN METERS? ☺

SUMMARY

AIR MOLECULES:

$$v_{\text{RMS}} = \sqrt{v^2} = \sqrt{\frac{3k_B T}{m}}$$

ROOT-MEAN-SQUARE "TYPICAL" SPEED (m/s) OF MOLECULES OR ATOMS (RANDOM DIRECTIONS!)
 (HOT PARTICLES = FASTER)
 MASS OF MOLECULES OR ATOMS (kg)
 TEMPERATURE (K = °C + 273)
 BOLZMANN: $1.4 \times 10^{-23} \text{ J/K}$

ACCELERATION & FORCE:

$$a = \frac{F}{m}$$

ACCELERATION (m/s²)
 FORCE (N = kg m/s²)
 MASS (kg)
 (HEAVY THINGS ARE HARD TO ACCELERATE)

IDEAL GAS:

$$P = \frac{N}{V} k_B T$$

PRESSURE (N/m²)
 PARTICLE #
 VOLUME
 (PRESSURE SCALES w/ PARTICLE DENSITY & TEMPERATURE)

TRANSVERSE WAVE, IDEAL STRING UNDER TENSION:

$$c = \sqrt{\frac{F_T}{\mu}}$$

WAVE SPEED (m/s)
 TENSION FORCE (N)
 LINEAR MASS DENSITY (kg/m)
 (TIGHT & LIGHT = FAST)

SOUND IN AIR:

$$c = 1.2 \sqrt{\frac{\bar{P}}{\rho}} = 1.2 \sqrt{\frac{k_B T}{m}}$$

MEAN PRESSURE N/m²
 DENSITY (kg/m³)
 PARTICLE MASS (kg)
 (HOT & LIGHT = FAST)
 1.2 FOR DIATOMIC
 1.3 FOR MONATOMIC

SOUND IN GENERAL:

$$c = \sqrt{\frac{1}{\rho \beta_s}}$$

ADIABATIC COMPRESSIBILITY (m²/N)
 (INCOMPRESSIBLE & LIGHT = FAST)